

- (b) (i) Determine the unstretched length of the rope.

length = m [1]

- (ii) Determine the spring constant k of the rope.

$k = \dots\dots\dots \text{Nm}^{-1}$ [2]

- (c) For the position of the bungee jumper at a distance of 120 m below the bridge:

- (i) show that the loss of gravitational potential energy since leaving the bridge is 75 kJ

[2]

- (ii) show that the elastic potential energy in the rope is 75 kJ.

[2]

- (d) Explain what can be deduced from the information in (c) about the speed of the bungee jumper when at a distance of 120 m below the bridge.

.....

 [2]

[Total: 10]

